

Spherepop Experiment Spec: Multi-Timescale Continuation

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1 Purpose

This experiment family operationalizes the bridge between *The Geometry of Boredom* and Spherepop by modeling boredom as a scope-selection problem over multiple unresolved histories with heterogeneous temporal horizons.

Remark 1.1 (Disambiguation from Appendix B Plan B). *This experiment suite is unrelated to the repository’s pending Appendix B poset-integration rewrite. It does not change Spherepop’s four primitives or claim that full Appendix B semantics is already implemented. It introduces a separate scheduling formalization for multi-timescale continuation using existing vocabulary.*

2 Core Formalization

Remark 2.1 (Grounding and status). *This document treats EMCG-based scope scheduling as a new proposed extension, not as an already-established theorem in the existing Spherepop specification. A related but distinct saturation notion already exists in this corpus via the Asymptotic Saturation Theorem in `textbook/The.Ecology.of.Distinctions.tex` (§“Asymptotic Saturation Without Exhaustion”), which concerns reintegration dynamics rather than task-portfolio scheduling. The present experiment asks whether an analogous local-vs-field saturation distinction is useful in practice at the workflow level.*

Definition 2.2 (Scope portfolio). *A portfolio is*

$$\mathcal{T} = \{(S_i, \tau_i)\}_{i=1}^n, \quad S_i = (O_i, \tau_i), \quad \tau_1 \ll \tau_2 \ll \dots \ll \tau_n,$$

where each unresolved scope S_i has local option space O_i and horizon τ_i .

Definition 2.3 (Local and field saturation). *With operator $F_t = C(H_t)$,*

$$\text{LocalSat}(i, t) : \max_{a \in O_i} \text{EMCG}_i(a \mid F_t) \approx 0,$$

$$\text{FieldSat}(t) : \max_i \max_{a \in O_i} \text{EMCG}_i(a \mid F_t) \approx 0.$$

Proposition 2.4 (Scope-switch recovery). *LocalSat(i, t) does not imply FieldSat(t). If another scope j satisfies $\max_{a \in O_j} \text{EMCG}_j(a \mid F_t) > 0$, switching scopes restores progress.*

Proposition 2.5 (Operator-drift recovery). *A scope can transition from $\text{EMCG}_i(a \mid F_t) \approx 0$ to $\text{EMCG}_i(a \mid F_{t+\Delta t}) > 0$ without changing O_i , due to intervening work on other scopes that modifies F .*

Proposition 2.6 (BIND-as-progress). *Long-horizon scopes can accumulate useful structure through BIND/REFUSE/COLLAPSE while remaining unresolved; progress does not require immediate POP.*

3 Scheduling Policies Under Comparison

1. **Novelty-only**: prefer longest-idle scope.
2. **Shortest-task-first**: always choose minimal horizon.
3. **Max-EMCG**: greedily maximize immediate expected marginal compression gain.
4. **EMCG/cost + anti-starvation**: maximize gain-per-cost with forced periodic updates to slow scopes.

4 Event Semantics in This Experiment Family

The run model interprets Spherepop operations as:

POP : resolve ready local scope,
REFUSE : remove continuation branch,
BIND : tighten continuation constraints,
COLLAPSE : merge distinctions no longer worth explicit tracking.

5 Metrics

- Time-to-first-local-saturation
- Time-in-field-saturation
- Mean recovery latency after scope switch
- Long-horizon starvation rate
- Compression-progress stability across horizons

6 Deliverables and Outputs

- Executable policy comparison script: `run.py`
- Results table template: `results-template.csv`
- Policy summaries including per-scope event counts and saturation statistics

7 Application Mapping

Personal workflow

Maintain active scopes at minute/day/week/year horizons.

Education

Alternate short and long-horizon work instead of novelty churn.

Research/programming

Keep deep threads unresolved while adding regular BIND-like constraints.

AI agents

Use multi-clock schedulers with anti-starvation guarantees.